

Effective group index of a 500-nm silicon waveguide

Ivo Straka
ivo.straka@gmail.com

Date: July 6, 2026

Abstract—The objective of this measurement is to determine the effective group index of a silicon-on-insulator integrated waveguide as a part of the edX UBCx *Phot1x Silicon Photonics Design, Fabrication and Data Analysis* course. Several Mach-Zehnder interferometers were designed, fabricated, and measured. The measured data are used to estimate the effective group index, and the results are compared to Lumerical MODE simulations.

I. INTEGRATED WAVEGUIDES

The waveguide is Si on SiO₂ with dimensions 500 × 220 nm. The target polarization mode is transverse-electric (TE) at the wavelength 1550 nm. Simulations were carried out in Lumerical MODE to determine the effective refractive index and the effective group index of the primary guided TE mode. The parameters of the simulation were: simulation area 5 × 5 μm and mesh grid of 500 × 500 samples. The E-field intensity drops by 18 orders of magnitude towards the edge of the simulation, which is set as a metal boundary. This should be sufficient for accurate modelling of the mode. Fig. 1 shows the field profile.

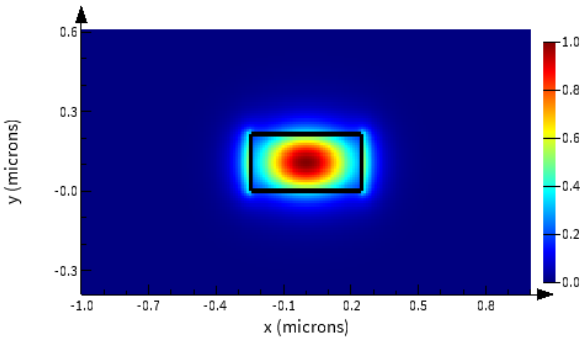


Fig. 1: The transversal profile of the electric field intensity across the 500 × 220 nm waveguide. A sub-area of the simulated region of 5 × 5 μm is shown. Simulated in Lumerical MODE.

The primary TE guided mode experiences an effective refractive index n_{eff} , and an effective group velocity index

$$n_g = n_{\text{eff}} - \frac{dn_{\text{eff}}}{d\lambda} \lambda. \quad (1)$$

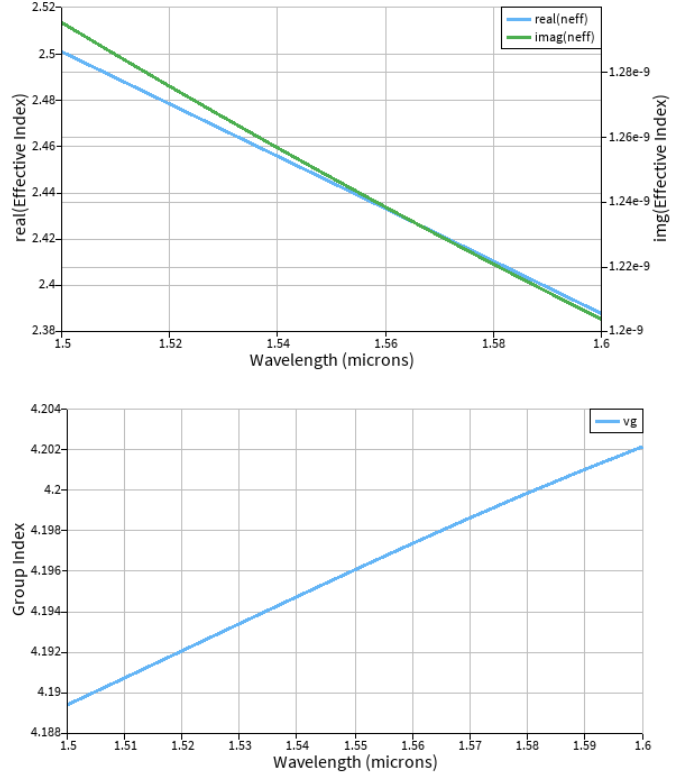


Fig. 2: Plots of simulated n_{eff} (top) and n_g (bottom). The imaginary part of n_{eff} represents optical loss. Simulated in Lumerical MODE.

The respective simulations are plotted in Fig. 2. The resulting compact model of the waveguide as a function of wavelength λ is

$$n_{\text{eff}}(\lambda) = 2.443 - 1.130 \mu\text{m}^{-1} \times (\lambda - 1.55 \mu\text{m}) - 0.04171 \mu\text{m}^{-2} \times (\lambda - 1.55 \mu\text{m})^2. \quad (2)$$

In order to account for variations in fabrication, a corner analysis was performed for variations in waveguide width and height. The refractive indices variations are listed in Table I.

II. PHOTONIC DEVICES AND METHODS

The whole layout is shown in Fig. 3. Five Mach-Zehnder interferometers (MZI1–5) were designed in *KLayout* using

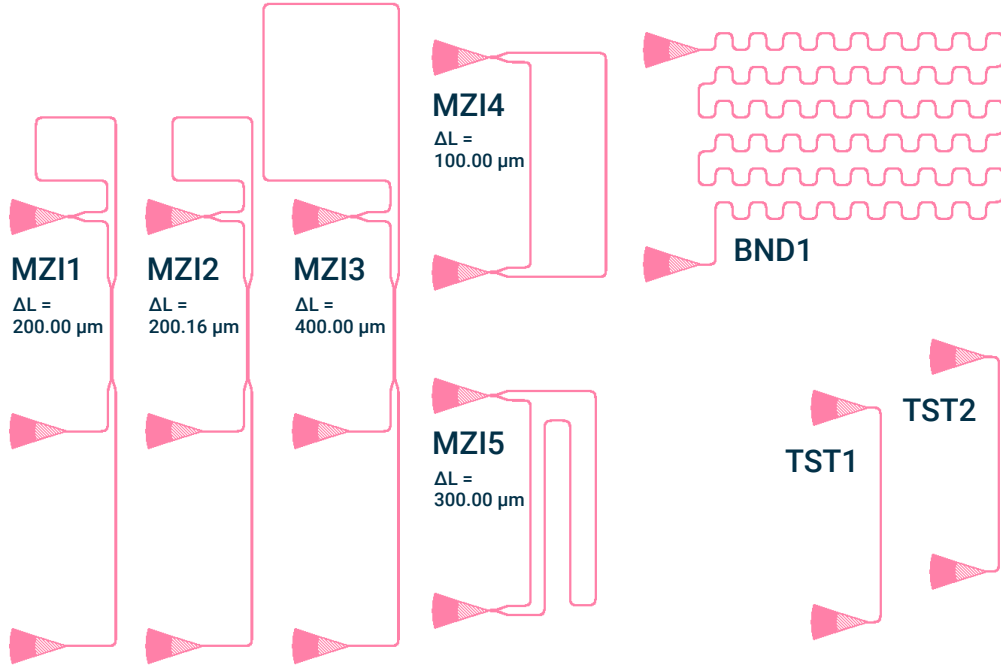


Fig. 3: The full layout shows five MZI structures, where MZI1–3 use a Y-branch and a broadband directional coupler, and MZI4–5 use two Y-branches. BND1 serves to estimate bending loss, and TST1–2 serve for de-embedding coupling loss. In each structure, the topmost grating coupler is the input, and the others are outputs. Designed in KLayout using EBeam and SiEPIC libraries.

width [nm]	height [nm]	n_{eff}	n_g
500	220.0	2.444	4.196
470	215.3	2.375	4.247
470	223.1	2.404	4.255
510	215.3	2.442	4.176
510	223.1	2.471	4.183

TABLE I: Corner analysis of the waveguide dimensions. Simulated in Lumerical MODE.

the *EBeam* and *SiEPIC* libraries. The TST1 and TST2 structures are used for de-embedding the insertion loss of the grating couplers, and examine the repeatability of this correction. Finally, the BND1 structure is meant to estimate the bending loss over 214 bends. All waveguides have width 500 μm , all bends were designed with a radius of 5 μm and Bézier parameter 0.2.

The MZIs were chosen to have various path length differences, with MZI2 chosen to have $\Delta L := 200 \mu\text{m} + \lambda / (4n_{\text{eff}})$ so that the interferometric extrema fall exactly between the extrema of MZI1 to enhance the sampling. In the end, the wavelength sampling was not a limiting factor, and the data sets were evaluated separately. All MZI parameters are listed in Table II.

Each MZI consists of an input 50:50 beam splitter, two arms with a path difference ΔL , and a second recombining 50:50 beam splitter [1]. The output optical power of the MZI is

$$I_{\text{out}} = I_0 \left(\frac{1}{2} \pm \frac{1}{2} V \cos \left(\frac{2\pi n_{\text{eff}} \Delta L}{\lambda} \right) \right). \quad (3)$$

The plus sign corresponds to the transmitted port, and the minus sign to the reflected port. For MZI4–5, there is only the transmitted output port available. The visibility $V \in [0, 1]$ and power term I_0 can both depend on λ . The term I_0 changes with the coupling efficiency in and out of the device, which is corrected by measuring the de-embedding structures TST1–2. Both I_0 and V determine the interference maxima and minima. These depend on the splitting ratio of the beam splitters. If both beam splitters deviate from 50:50 splitting ratio in the same way, only the interference minimum will be affected. In a more general case, such as for MZI1–3 that feature different beam splitters, both maxima and minima are going to be affected.

A simulation of MZI1 is shown in Fig. 4. The plots show a characteristic interference pattern of Eq. (3) in log-scale.

We are going to be scanning the output power as a function of λ . The distance between two neighboring interference minima at λ and λ' on either output port are bound by the periodicity of the interference pattern:

$$\frac{n_{\text{eff}}(\lambda) \Delta L}{\lambda} - \frac{n_{\text{eff}}(\lambda') \Delta L}{\lambda'} = 1. \quad (4)$$

We are going to assume that their distance is small relative to the wavelength λ and the curvature of n_{eff} , and so $\lambda' = \lambda + d\lambda$. A first-order infinitesimal expansion of the

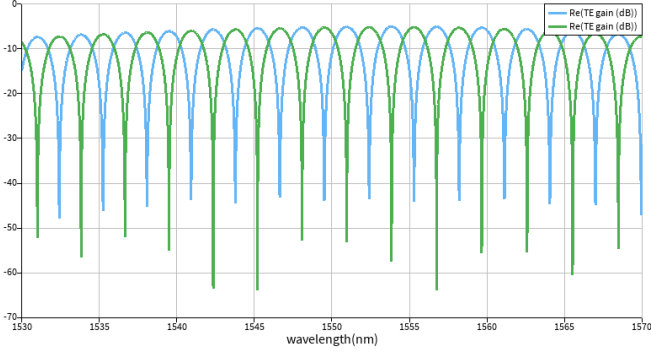


Fig. 4: A simulation of transmission of both MZI1 output ports. Simulated in Lumerical INTERCONNECT.

above terms leads to

$$\frac{n_{\text{eff}}}{\lambda} - \left(n_{\text{eff}} + \frac{dn_{\text{eff}}}{d\lambda} \lambda \right) \left(\frac{1}{\lambda} - \frac{d\lambda}{\lambda^2} \right) = \frac{1}{\Delta L}, \quad (5)$$

$$\left(n_{\text{eff}} - \frac{dn_{\text{eff}}}{d\lambda} \lambda \right) d\lambda = \frac{\lambda^2}{\Delta L}. \quad (6)$$

The distance $d\lambda$ between two interference minima is called the free spectral range (FSR), and the term on the left side is the group index from Eq. (1). This leads to the expression

$$n_g(\lambda) \approx \frac{\lambda^2}{\text{FSR}(\lambda) \times \Delta L}. \quad (7)$$

Here, the FSR will be extracted from the experimental data for each MZI as a function of λ , and the results will be fit to obtain the group index. This method is preferred over fitting the whole wavelength sweep, because a least-squares fit relies on independent statistical errors, whereas the wavelength sweeps are affected by several systematic errors. These include uncorrected background insertion loss and varying visibility profile. The locations of the interference minima are affected very little by these effects, as long as they vary slowly.

III. RESULTS

The data were taken as a wavelength sweep around 1550 nm with TE polarization at chip temperature of 30 °C. The scanning resolution was 0.008 nm and to minimize the influence of measurement artifacts, a range of 1530–1570 nm was chosen for data processing.

First, the power transmitted through the de-embedding structures TST1–2 is fit with a third-degree polynomial. The results are shown in Fig. 5. The two profiles are different, despite the structures being next to each other. The measurement is probably limited by the repeatability of the fiber alignment.

The bending loss was tested on BND1 with the de-embedded data shown in Fig. 6. The only region, where de-embedding using both structures gives similar results, is around 1550 nm. If we accept that the total insertion loss is ≤ 0.9 dB at 1550 nm, and considering that the simulated straight-waveguide loss for the corresponding length should be negligible (10^{-4} dB), then one bend should have a

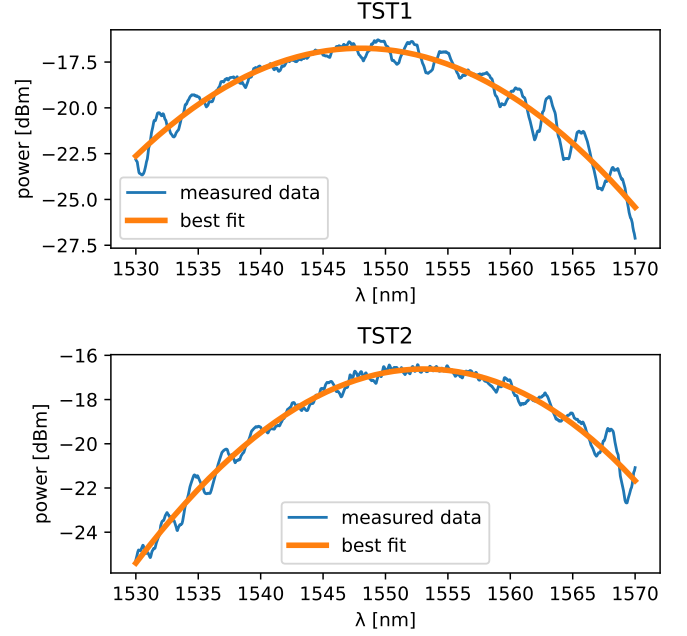


Fig. 5: De-embedding coupling loss. Fit with a third-degree polynomial.

bending loss ≤ 0.004 dB. This result is, however, extremely uncertain.

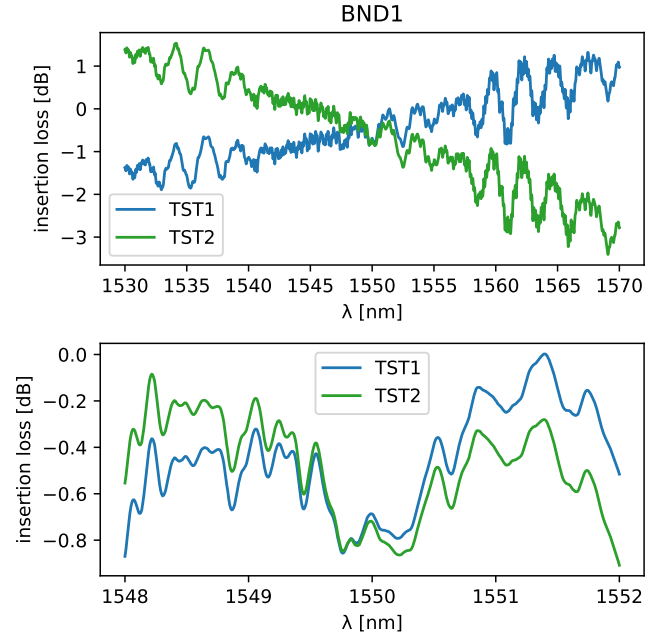


Fig. 6: Bending loss measurement after de-embedding using both TST structures. The bottom zoom-in shows a region around 1550 nm.

The results of the MZI sweeps are shown in Fig. 7. For every MZI, a suitable de-embedding structure was chosen to minimize the background slope. The interference minima were found using a SciPy peak-finding algorithm.

structure	ΔL	FSR (theor.)	FSR (sim.)	FSR (meas.)	n_g (sim. circ.)	n_g (meas.)
MZI1	200.00 μm	2.86 nm	2.85 nm	2.82 nm	4.181(5)	4.18(1)
MZI2	200.16 μm	2.86 nm	2.88 nm	2.90 nm	4.186(5)	4.18(1)
MZI3	400.00 μm	1.43 nm	1.43 nm	1.44 nm	4.185(5)	4.183(5)
MZI4	100.00 μm	5.76 nm	5.59 nm	5.74 nm	4.25(2)	4.196(1)
MZI5	300.00 μm	1.91 nm	1.90 nm	1.92 nm	4.200(4)	4.195(6)

TABLE II: The MZIs showing the path length differences and the corresponding FSR values that are nearest 1550 nm. Shown are the theoretical values calculated using the simulated n_g , the values simulated in Lumerical INTERCONNECT, and the values measured. The resulting best-fit values of the group index are shown for circuit simulations and measurements in the last two respective columns.

The correspondence of the measurement channels to the grating couplers in the layout is uncertain, because according to the course specifications, the live channels for MZI1–3 should be 1 and 2, but in the data files, the live channels are 2 and 3. This, however, does not significantly affect the results of our measurement. We estimate the FSR using distances between the neighboring peaks, which in case of MZI1–3 are equal to $\text{FSR}/2$ due to both output channels being evaluated. Each FSR sample is assigned to a wavelength that is the midpoint between the two respective minima.

The FSR samples are inserted in Eq. 7 to estimate the group index at 1550 nm using a least-squares fit, as shown in Fig. 8. All data within one measurement are assumed to have a normally distributed independent errors of identical, but unknown variance. To accommodate for potential group index dispersion, a linear slope of n_g was added. The statistical significance of the slope parameter is, however, low for all results (lowest adjusted p -value 0.64).

The experimentally obtained values are listed in Table II. The FSR values at 1550 nm are theoretically calculated using Eq. (7) and the simulated value of $n_g = 4.196$. Then, the FSR simulated in Lumerical INTERCONNECT for the designed MZI circuits are listed. The simulated values exhibits variations from the theoretical values, as the measured values do, but the spread differs. For example, for MZI4, the simulated FSR values deviate much more in the simulation than they do in the experimental data. The reason for the deviations from the expected trend could be variations in the numerical inputs to the simulation, but also some systematic effect in the design. However, this systematic effect is not present in the experimental data at the same magnitude, which makes this hypothesis unlikely. Without detailed knowledge of the simulation, the reason cannot be determined.

The maximum-likelihood estimates for n_g at 1550 nm are shown in Table II and plotted in Fig. 9 (top). All the results lie within the corner analysis. MZI1–3 exhibit systematically lower values of the group index than the simulated value, while MZI4–5 match the simulation perfectly. The main factor seems to be the MZI geometry rather than the position of the structures within the layout. For comparison, the FSR values were simulated in Lumerical INTERCONNECT for the designed circuits, as shown in the lower plot of Fig. 9. Apart from very large variations in the MZI4 simulation, the group index extracted from MZI1–3

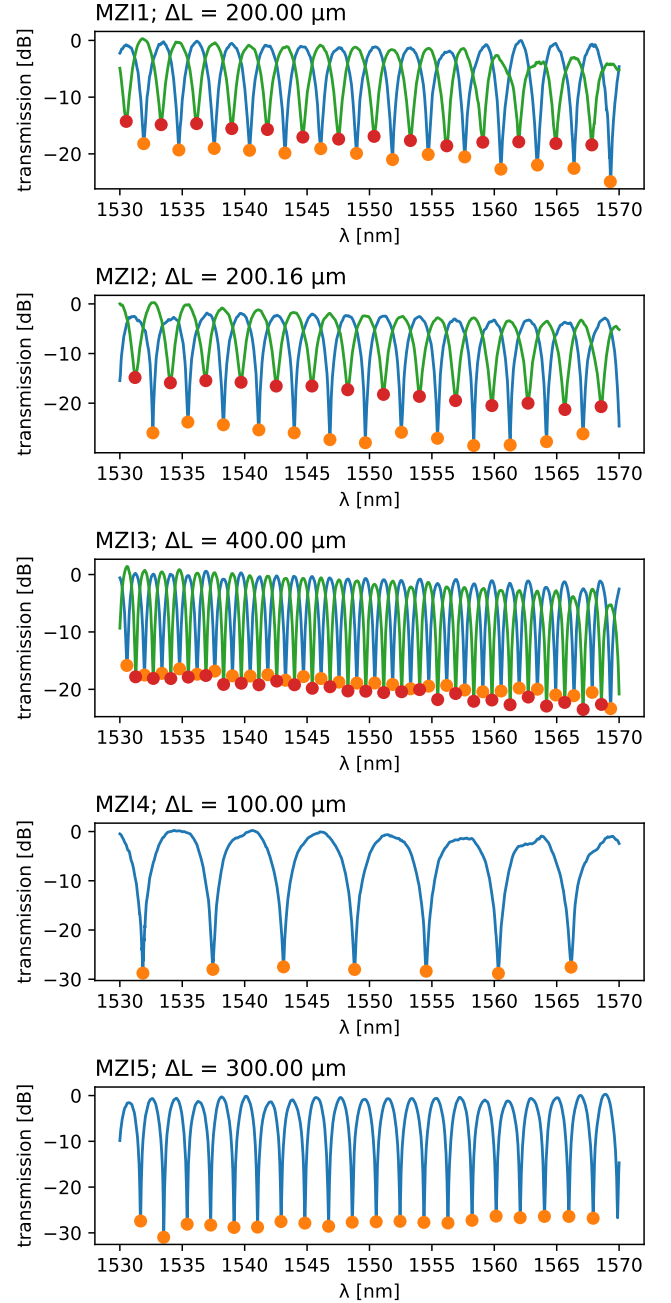


Fig. 7: The transmission of every MZI structure after de-embedding for coupling loss. Interference minima are highlighted for every channel.

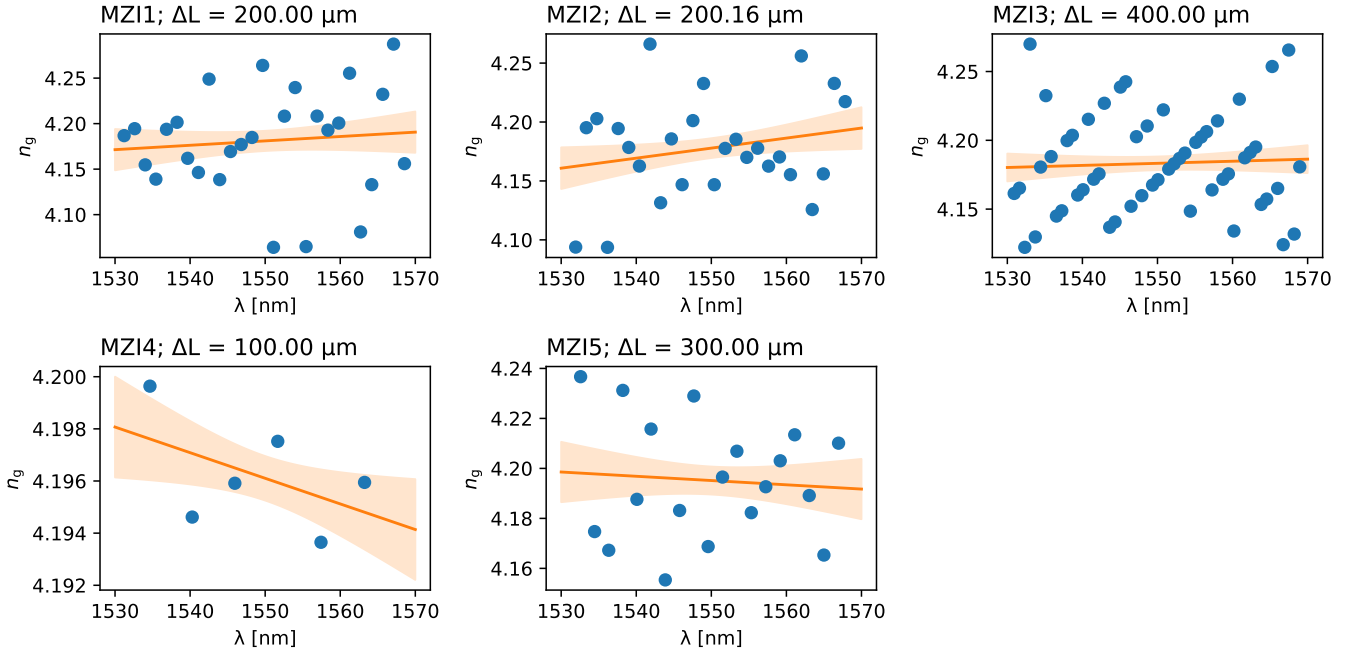


Fig. 8: The group indexes extracted from the measured FSRs, and corresponding least-square fits. The data pattern in MZI3 is caused by a low FSR, where the values vary by discrete steps due to finite wavelength resolution.

also exhibits systematically lower values than MZI4–5. This is caused by the FSRs being systematically higher than modelled. Due to the amount of bends and path differences being comparable between MZI5 and MZI1 and MZI3, the likely explanation is an artefact imposed by the use of the broad-band splitter at the output.

IV. CONCLUSION

Several MZI geometries were designed to estimate the group index of a Si $500 \times 220 \mu\text{m}$ waveguide. Path differences between $100 \mu\text{m}$ and $400 \mu\text{m}$ were used. Coupling losses were satisfactorily de-embedded using dedicated structures, and the measured FSR were used to estimate the group index. The experimental results are within 0.4% of the simulated value of $n_g = 4.196$ at 1550 nm . There is a slight systematic variation that depends on the type of beam splitter used in the MZI, which was supported by circuit simulations.

APPENDIX

FABRICATION AND MEASUREMENTS

A. Applied Nanotools, Inc. NanoSOI process

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (<http://www.appliednt.com/nanosoi>; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and $2 \mu\text{m}$ buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of $25 \times 25 \text{ mm}$, and lines were scribed into the substrate

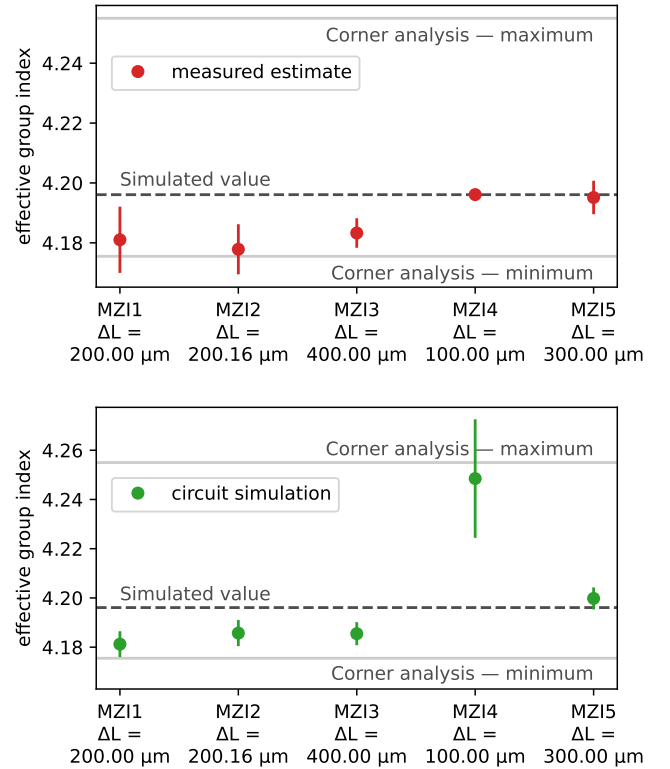


Fig. 9: The resulting estimates of n_g for each MZI. The top graph shows experimentally obtained estimates, and the bottom graph shows the circuit simulation results from Lumerical INTERCONNECT, obtained by fitting the simulated FSR values.

backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H_2SO_4 : H_2O_2) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 μm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300 °C. Reflectometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

B. Measurement description

To characterize the devices, a custom-built automated test setup [3], [7] with automated control software written in Python was used [4]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [5]. A 90° rotation was used to inject light into the TM grating couplers [5]. A polarization maintaining fibre array was used to couple light in/out of the chip [6].

C. Acknowledgments

I acknowledge the edX UBCx Phot1x Silicon Photonics Design, Fabrication and Data Analysis course, which is supported by the Natural Sciences and Engineering Research Council of Canada (NSERC) Silicon Electronic-Photonic Integrated Circuits (SiEPIC) Program. The devices were fabricated by Richard Bojko at the University of Washington Washington Nanofabrication Facility, part of the National Science Foundation's National Nanotechnology Infrastructure Network (NNIN), and Cameron Horvath at Applied Nanotools, Inc. Mateo Branion-Calles performed the measurements at The University of British Columbia. We acknowledge Lumerical Solutions, Inc., Mathworks, Mentor Graphics, Python, and KLayout for the design software.

REFERENCES

- [1] B. E. A. Saleh and M. C. Teich, *Fundamentals of Photonics*, Second Edition, Wiley (2007).
- [2] R. J. Bojko, J. Li, L. He, T. Baehr-Jones, M. Hochberg, and Y. Aida, "Electron beam lithography writing strategies for low loss, high confinement silicon optical waveguides," *J. Vacuum Sci. Technol. B* 29, 06F309 (2011)
- [3] Lukas Chrostowski, Michael Hochberg, chapter 12 in "Silicon Photonics Design: From Devices to Systems", Cambridge University Press, 2015
- [4] <http://siepic.ubc.ca/probestation>, using Python code developed by Michael Caverley.
- [5] Yun Wang, Xu Wang, Jonas Flueckiger, Han Yun, Wei Shi, Richard Bojko, Nicolas A. F. Jaeger, Lukas Chrostowski, "Focusing sub-wavelength grating couplers with low back reflections for rapid prototyping of silicon photonic circuits", *Optics Express* Vol. 22, Issue 17, pp. 20652-20662 (2014) doi: 10.1364/OE.22.020652
- [6] www.plcconnections.com, PLC Connections, Columbus OH, USA.
- [7] <http://mapleleafphotonics.com>, Maple Leaf Photonics, Seattle WA, USA.